Def. A Group G is finitely generated if: there is a finite subset A of G s.t. $G = \langle A \rangle$.

Def. For each $r \in \mathbb{Z}$ with $r \geq 0$, $\mathbb{Z}^r = \overbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}^{r}$ is called free abelian group of rank r.

Fundamental Theorem of Finitely Generated Abelian Group.
Let G be a finitely generated abelian group, Then

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_s}$$

for some integers $r, n_1, \cdots, n_s$ satisfying:
$r \geq 0$, $n_j \geq 2$ for all $1 \leq j \leq s$, and $n_{i+1} \mid n_i$ for $1 \leq i \leq s-1$.
And, such expression is unique.

Corollary. If $n \in \mathbb{N}$ is the product of distinct primes,
then up to isomorphism, the only abelian group of order n is the cyclic group of order n.

Thm. Let G be an abelian group of order $n > 1$ and let

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$$

be the prime decomposition. Then,
1). $G \cong A_1 \times A_2 \times \cdots \times A_k$ where $|A_i| = p_i^{\alpha_i}$.
2). For each $A \in \{A_1, \cdots, A_k\}$ with $|A| = p^{\alpha}$,

$$A \cong \mathbb{Z}_{p^{\beta_1}} \times \cdots \times \mathbb{Z}_{p^{\beta_t}} \text{ with } \beta_1 \geq \cdots \geq \beta_t \geq 1, \ \beta_1 + \cdots + \beta_t = \alpha.$$

where t and $\beta_1, \cdots, \beta_t$ depend on i.
Moreover, such expression is unique.

Lemma. Let $m, n \in \mathbb{N}$. Then,

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \text{ if and only if } (m,n) = 1.$$

Proof. Denote $\mathbb{Z}_m = \langle x \rangle$, $\mathbb{Z}_n = \langle y \rangle$. Since $mn = \operatorname{lcm}(m,n) \cdot \gcd(m,n)$,

$$\operatorname{lcm}(m,n) = mn \quad \text{iff} \quad \gcd(m,n) = 1.$$

And, for any $x^a \in \langle x \rangle$ and $y^b \in \langle y \rangle$,

$$(x^a y^b)^{\operatorname{lcm}(m,n)} = x^{a \cdot \operatorname{lcm}(m,n)} \cdot y^{b \cdot \operatorname{lcm}(m,n)} = 1 \cdot 1 = 1.$$

If $(m,n) \neq 1$, then $\operatorname{lcm}(m,n) = (mn)/\gcd(m,n) < mn$, so every element in $\mathbb{Z}_m \times \mathbb{Z}_n$ has an order smaller than mn, so $\mathbb{Z}_m \times \mathbb{Z}_n \ncong \mathbb{Z}_{mn}$.

If $(m,n) = 1$, then $|xy| = mn$, so (x,y) generates $\mathbb{Z}_{mn}$, cyclic.